SENTHIL PUBLIC SCHOOL, DHARMAPURI

AFFILIATED TO CBSE, NEW DELHI (AFFL. NO. 1930248)

CLASS: CLASS 8 (CBSE)	**SUBJECT: MATHEMATICS**	**DATE: 18-OCT-2026**
TIME: 45 MINS	**PHASE: Phase 3: Morning Exam Sprint (Mathematics Rapid Recall)**	**MAX. MARKS: 25**

--- SECTION A — OBJECTIVE & ASSERTION-REASON ---

Q.1 [MTH-P3-Q01] Which of the following algebraic identities is universally correct for all real values of a and b? **[1]**
(a) $(a + b)^2 = a^2 + b^2$ (b) $(a - b)^2 = a^2 - 2ab + b^2$ (c) $(a + b)(a - b) = a^2 + b^2$ (d) $(a + b)^2 = a^2 - 2ab + b^2$

Q.2 [MTH-P3-Q02] The coefficient of x^2 in the expanded form of $(3x - 4)^2$ is: **[1]**
(a) 9 (b) -24 (c) 16 (d) -9

Q.3 [MTH-P3-Q03] The value of $(-2/3)^4$ expressed as a positive rational number is: **[1]**
(a) 16/81 (b) -16/81 (c) 8/27 (d) -8/27

Q.4 [MTH-P3-Q04] Which of the following algebraic identities is universally correct for all real values of a and b? **[1]**
(a) $(a + b)^2 = a^2 + b^2$ (b) $(a - b)^2 = a^2 - 2ab + b^2$ (c) $(a + b)(a - b) = a^2 + b^2$ (d) $(a + b)^2 = a^2 - 2ab + b^2$

Q.5 [MTH-P3-Q05] The coefficient of x^2 in the expanded form of $(3x - 4)^2$ is: **[1]**
(a) 9 (b) -24 (c) 16 (d) -9

--- SECTION B — VERY SHORT CONCEPT DRILL ---

Q.6 [MTH-P3-Q06] Factorise the algebraic expression completely by grouping terms: $15xy - 6x + 5y - 2$. **[2]**

Q.7 [MTH-P3-Q07] Solve the linear equation for y: $5y - 7 = 2y + 8$ and verify your solution step-by-step. **[2]**

--- SECTION C — NUMERICAL & CONCEPT APPLICATION ---

Q.8 [MTH-P3-Q08] A cuboid has dimensions 60 cm x 54 cm x 30 cm. How many small cubes with side 6 cm can be placed in the given cuboid? **[3]**

Q.9 [MTH-P3-Q09] The perimeter of a rectangular swimming pool is 154 m. Its length is 2 m more than twice its breadth. Find the length and breadth of the pool. **[3]**

--- SECTION D — DERIVATION & APPLICATION TASK ---

Q.10 [MTH-P3-Q10] (a) A closed cylindrical storage tank of radius 7 m and height 3 m is made from a sheet of metal. How much sheet of metal is required? (Take $\pi = 22/7$). (b) If the metal costs ☐ 45 per square meter, calculate the total cost. **[5]**

Q.11 [MTH-P3-Q11] A road roller takes 750 complete revolutions to move once over to level a road. Find the area of the road if the diameter of the road roller is 84 cm and length is 1 m. (Take $\pi = 22/7$). **[5]**